Skills Summary

Radicals Do Not Split Across a Sum

Use this reference while completing your worksheet or when studying.

Skill 1 — Test a suspect rule with numbers

Method: one counterexample kills a proposed identity. Pick two easy numbers, evaluate both sides, and compare. If they differ even once, the rule is false — no further argument is needed.

Example: Is $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$?
Step 1. A claimed identity has to hold for every choice, so the cheapest test is to try one pair of perfect squares and compare the two sides.
Step 2. Take $a = 9$ and $b = 16$: the left side is $\sqrt{25}$, and the right side is $3 + 4$.
left side **5**, right side **7** — so the rule is false.

Try it: Test the same claim with $a = 36$ and $b = 64$.

check: left side **10**, right side **14**

Skill 2 — Roots split over products, not sums

The rules that do hold (for $a, b \geq 0$): $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$ and $\sqrt{\dfrac{a}{b}} = \dfrac{\sqrt{a}}{\sqrt{b}}$. There is no rule for $\sqrt{a+b}$. To simplify a radical, look for a perfect-square *factor*.

Example: Simplify $\sqrt{48}$.
Step 1. Simplifying means pulling out a perfect square, and that is legal only because 48 is being written as a product, not as a sum.
Step 2. $48 = 16 \cdot 3$, so $\sqrt{48} = \sqrt{16}\sqrt{3} = 4\sqrt{3}$.
$\Rightarrow$ $\mathbf{4\sqrt{3}}$

Try it: Simplify $\sqrt{98}$.

check: $\mathbf{7\sqrt{2}}$

Skill 3 — Spotting the split-over-a-sum error

Warning sign: a step that turns $\sqrt{\text{something} + \text{something}}$ into two separate roots, or turns $\sqrt{A} + \sqrt{B}$ into one root. Both are the same illegal move. A sum of two squares such as $x^2 + 16$ does not factor over the real numbers, so its square root is already in simplest form.

Example: A student writes $\sqrt{x^2 + 16} = x + 4$. Test it at $x = 6$.
Step 1. The claim is an identity, so substituting one convenient value and comparing the two sides is enough to accept or reject it.
Step 2. Left side: $\sqrt{36 + 16} = \sqrt{52} = 7.2111\ldots$; right side: $6 + 4$.
left side **7.21**, right side **10** — the claim fails.

Try it: Test the claim $\sqrt{x^2 + 25} = x + 5$ at $x = 12$.

check: left side **13**, right side **17**